

Travel & Tourism Booking Analytics

Power BI Data Analytics Project — Problem Statement & Project Blueprint

1. Background

The travel and tourism industry generates large volumes of booking data across destinations, customers, travel dates, booking values, discounts, ratings, loyalty memberships, and cancellations. Raw booking data can contain missing values, duplicate records, and inconsistent formats. A centralized Power BI solution can transform this data into actionable insights for management.

2. Problem Statement

A travel booking platform has collected 50,000+ raw booking records covering multiple years but lacks a centralized reporting system. Management has difficulty monitoring booking trends, revenue performance, customer behavior, destination performance, and cancellation patterns.

Key management questions

- Which destinations generate the highest revenue?
- How are bookings trending over time?
- What is the total and average booking revenue?
- What percentage of bookings are cancelled?
- Which destinations have the highest cancellation rates?
- How does customer loyalty affect booking behavior and revenue?
- Which travel types are most popular?
- What is the average customer rating?
- How much discount is provided?
- Which destinations and travel categories perform best?

3. Objective of the Project

- Clean and standardize the raw booking data using Power Query.
- Handle missing values and duplicate records.
- Correct inconsistent data types and formats.
- Build an appropriate data model.
- Create business KPIs using DAX.
- Analyze booking and revenue trends.
- Identify high-performing destinations.
- Analyze cancellation patterns.
- Understand customer loyalty and behavior.
- Build an interactive multi-page Power BI dashboard.

4. Scope of the Project

Area	Analysis
Bookings	Total and monthly bookings
Revenue	Total, average and destination revenue
Customers	Travelers, ratings and loyalty

Area	Analysis
Destinations	Revenue, bookings and ratings
Cancellations	Cancellation count and rate
Discounts	Discount amount and percentage
Travel	Trip duration and travel type
Time	Monthly/yearly trends

5. Key Business Questions

1. Total bookings?
2. Total revenue?
3. Average booking value?
4. Booking trend over time?
5. Revenue trend over time?
6. Top revenue destinations?
7. Top destinations by bookings?
8. Cancellation rate?
9. Destinations with highest cancellation?
10. Total travelers?
11. Most popular travel types?
12. Average customer rating?
13. Effect of loyalty membership?
14. Total discount?
15. Best-rated destinations?
16. Average trip duration?
17. Revenue by customer segment?
18. Booking value by destination?
19. Highest-value bookings?
20. Which areas require management attention?

6. Power BI Dashboard Pages

Page 1 — Executive Summary

KPI cards: Total Bookings, Total Revenue, Total Travelers, Average Booking Value, Cancellation Rate, Average Rating.
Charts: Monthly Booking Trend, Monthly Revenue Trend, Top 10 Destinations, Booking Status.

Page 2 — Booking & Revenue Analysis

Bookings by Month, Revenue by Month, Revenue by Destination, Bookings by Travel Type, Average Booking Value by Destination, Revenue by Loyalty Status. Slicers: Date, Destination, Travel Type, Loyalty Status.

Page 3 — Customer Analysis

Loyalty Members vs Non-Members, Average Rating by Destination, Rating Distribution, Travelers by Travel Type, Revenue by Loyalty Status, Cancellation Rate by Loyalty Status.

Page 4 — Destination Performance

Top 10 Destinations by Revenue, Top 10 Destinations by Bookings, Cancellation Rate by Destination, Average Rating by Destination, Average Trip Duration by Destination.

Page 5 — Cancellation & Discount Analysis

Cancelled vs Completed Bookings, Cancellation Trend, Cancellation Rate by Destination, Total Discount by Destination, Discount vs Booking Value, Revenue After Discount.

7. DAX KPI Categories

- Booking KPIs: Total Bookings, Completed Bookings, Cancelled Bookings, Cancellation Rate.
- Revenue KPIs: Total Revenue, Average Booking Value, Revenue per Traveler, Total Discount, Average Discount.
- Customer KPIs: Total Travelers, Average Rating, Loyalty Members, Loyalty Rate.
- Destination KPIs: Destination Revenue, Destination Bookings, Destination Cancellation Rate, Destination Rating.
- Time Intelligence: Monthly Bookings, Monthly Revenue, Previous Year Revenue, YoY Revenue Growth, YTD Revenue.

8. Data Cleaning Process

- Import raw CSV into Power BI.
- Open Power Query Editor.
- Remove duplicate booking records.
- Identify and handle null or missing values.
- Correct column names.
- Set appropriate data types.
- Standardize inconsistent text values.
- Check invalid values and potential outliers.
- Create required derived columns.
- Validate the cleaned dataset before visualization.

9. Project Workflow

Raw Data → Power Query Cleaning → Data Validation → Data Model → DAX Measures → Interactive Dashboard → Business Insights → Management Recommendations

10. Expected Outcome

A multi-page Power BI report that transforms raw travel booking records into management-ready insights and enables filtering by date, destination, travel type, and loyalty status.

11. Tools & Technologies

- Data Source: CSV
- Data Transformation: Power Query / M
- Data Modeling: Power BI
- Calculations: DAX
- Visualization: Power BI cards, charts, matrices, slicers and maps where appropriate